Temporal Hand Labanotation: Upgrading Hand Notation into an Editable Motion Representation

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Abstract

Hand Labanotation (HL) established that hand motion can be written symbolically, but its original formulation still records motion largely frame by frame. Editing, retrieval, transfer, and interaction-aware reuse require more than isolated poses: they require short transitions and local coordination to be part of the representation itself. We present Temporal Hand Labanotation (Temporal HL), a temporal upgrade of HL that represents a motion segment with two coupled symbolic components: grouped anatomy-aligned state factors and short transition motifs over local runs. A deterministic family-repair operator stabilizes only within-family near-confusions, so the representation remains explicit and symbolic rather than absorbed into a latent temporal head. We evaluate Temporal HL on held-out unseen-subject splits with matched operator strength across structure, locality, and interaction-conditioned preservation audits. Grouped symbolic factors improve sequence accuracy over a flat symbolic baseline by +0.2051, while weighted symbolic locality reaches 0.3553 against learned proxy representations below 0.10. On hard interaction-conditioned preservation, corrected support relaxation raises full-slice joint success from 0.0250 to 0.1807 for closing and from 0.0533 to 0.1812 for opening; on broader feasible transfer, chunk-level temporal edits further improve closing from 0.7025 to 0.7785 and opening from 0.6306 to 0.7197. These results show that temporal structure belongs inside the notation itself.

Introduction

Hand motion needs a representation when the goal is not only to estimate pose, but also to document, edit, retrieve, and reuse motion structure. This matters for communication analysis, animation, robotics, and human-AI interfaces, where a useful hand representation must expose motion units that humans can inspect and operators can modify. Dominant hand representations such as keypoints, meshes, and parametric models remain precise but are not naturally symbolic or locally editable (Romero, Tzionas, and Black 2017); 2D/3D hand pipelines also remain vulnerable to occlusion and interaction complexity (Cao et al. 2021; Moon et al. 2020); and even large recent bimanual datasets still describe hand behavior primarily through geometric or textual channels rather than a reusable symbolic motion object (Fu et al. 2025). Recent bimanual synthesis and motion-language systems further reinforce that interaction-rich motion is now treated as a sequence-level object (Chen et al. 2025; Zhang et al. 2025). Hand Labanotation (HL) showed that symbolic hand documentation is viable at scale (Li et al. 2024), but its original formulation remains largely framewise. It captures pose symbolically, yet does not make short temporal units first-class objects for editing, retrieval, and transfer.

Temporal structure needs to be encoded in the notation itself. If temporal information is injected only through a latent sequence encoder, prediction may improve, but the representation still lacks explicit motion units that can be read, edited, and transferred. A framewise notation remains an archive of poses, not yet a symbolic motion object. We therefore upgrade HL into Temporal Hand Labanotation (Temporal HL), a transition-aware symbolic representation that stores a motion segment with grouped anatomy-aligned states, short transition motifs over local runs, and a deterministic family-repair operator that restores within-family consistency without changing symbolic intent.

At the object level, the distinction from old HL is simple. Old HL stores a frame t as a flat symbolic state vector over local hand regions. Temporal HL stores a motion segment as a pair (G, M), where G is a sequence of grouped symbolic states aligned with finger families and hand-level structure, and M is a sequence of short transition motifs attached to contiguous local runs. A deterministic family-repair map acts on (G, M) to enforce within-family consistency, but it is an operator on the representation rather than a third symbol type. The result is an explicit and editable segment object rather than a flat per-frame archive, while preserving the original HL inventory.

Temporal HL yields three representation-level gains. Grouped symbolic factors improve sequence accuracy over a flat symbolic baseline by +0.2051. Weighted symbolic locality reaches 0.3553, compared with 0.0945 and 0.0848 for matched semantic and continuous proxies. Under interaction, corrected support relaxation raises hard full-slice preservation from 0.0250 to 0.1807, while chunk transfer lifts broader feasible closing from 0.7025 to 0.7785. This evidence places temporal structure inside the representation itself and makes the notation more usable as an intermediate object for controllable hand-motion systems. All three gains are measured on held-out unseen-subject evaluation with matched downstream operator strength.

This framing also determines how the work should be evaluated. A representation paper should not be judged only by whether a decoder predicts frame labels more accurately; it should also be judged by whether the notation supports the operations for which such a representation is introduced in the first place: organizing state, keeping edits local, and preserving motion under downstream use.

Our contributions are:

- We define Temporal HL, a temporal symbolic hand-motion object that augments HL with anatomy-aligned grouped states, short transition motifs, and deterministic family repair.
- We show that this minimal temporal upgrade improves the properties a representation should expose: structure, local editability, and interaction-conditioned transfer.
- We instantiate held-out, unseen-subject evaluation protocols across three matched operator families so that the notation is tested as a representation object rather than only as the output space of a recognizer.

Related Work

Geometric hand representations. Current hand representations are dominated by keypoints, meshes, and parametric models. InterHand2.6M and FreiHAND remain standard benchmarks for 3D hand pose and shape (Moon et al. 2020; Zimmermann et al. 2019), and MANO provides a compact parametric hand model (Romero, Tzionas, and Black 2017). These representations are effective for estimation and reconstruction, but they remain geometric objects rather than symbolic motion units. Our goal is not to replace them for pose recovery, but to provide a symbolic layer for motion structure that geometric outputs do not directly expose.

Symbolic notation for movement. HL introduced a symbolic vocabulary, dataset, and recognition pipeline for hand movement documentation (Li et al. 2024). More recent notation-driven systems such as LabanLab emphasize that symbolic motion representations are used not only for archival recording but also for preview, editing, and downstream generation (Yan, Yu, and Zhou 2025). Our work builds on that notation line, but changes the central question. Instead of translating images into per-frame symbols more accurately, we ask what the notation object itself should contain if it is meant to represent hand motion rather than only hand pose.

Motion representations with temporal semantics. Recent motion-language work increasingly treats motion as a structured sequence-level object rather than a stack of independent frames (Chen et al. 2025). Large bimanual datasets and synthesis systems further reinforce that interaction and transition structure are central to hand behavior (Fu et al. 2025; Zhang et al. 2025). In parallel, notation-driven interfaces such as LabanLab show that symbolic motion objects are useful not only for archival recording but also for preview, editing, and downstream use (Yan, Yu, and Zhou 2025). Our setting is narrower and more symbolic: we do not seek a foundation model for motion generation, but a compact notation-level representation whose temporal structure is explicit, inspectable, and locally editable. This is also the distinction from simply attaching a stronger temporal encoder to framewise symbols: encoder-side temporalization may help prediction, but it does not by itself yield a symbolic object that can be read, edited, or transferred.

Representation Formulation

Object Definition

We formalize Temporal HL as a symbolic representation for short motion segments rather than isolated frames. Let $x_t = (s_t^1, \ldots, s_t^R)$ denote the old-HL description of frame t, where each s_t^r is the symbol assigned to one local hand region and R is the number of recorded regions. This frame object says what the hand state is at time t, but not how nearby states are coupled over time.

Temporal HL replaces this flat frame object with a temporal symbolic object $(G_{t:t+\ell-1}, M_{t:t+\ell-1})$ over a local motion segment of length ℓ. The design follows three representation desiderata: anatomy-aligned structure, explicit short-range temporal units, and symbolic consistency under local editing and transfer. The upgrade is additive rather than destructive: flattening the grouped state component recovers the old per-frame HL symbol inventory, while the motif component adds the temporal information that framewise HL does not encode. The two components serve these goals directly.

Grouped symbolic factors. For each frame, the state component $G_t = (g_t^1, \ldots, g_t^F, g_t^{\text{palm}})$ stores grouped symbols rather than a flat list. The groups are anatomy-aware: the four fingers, thumb, and hand-level structure are recorded separately so that local symbolic dependence matches hand anatomy. The goal is not compression; it is to preserve interpretable local independence.

Transition chunks. For a contiguous run $t{:}t{+}\ell{-}1$, the motif component $M_{t:t+\ell-1}$ stores a short symbolic transition aligned with the same grouped states. It records how the grouped symbols evolve over the run, so downstream retrieval and intervention operate on motion units rather than isolated poses. Concretely, the mo-

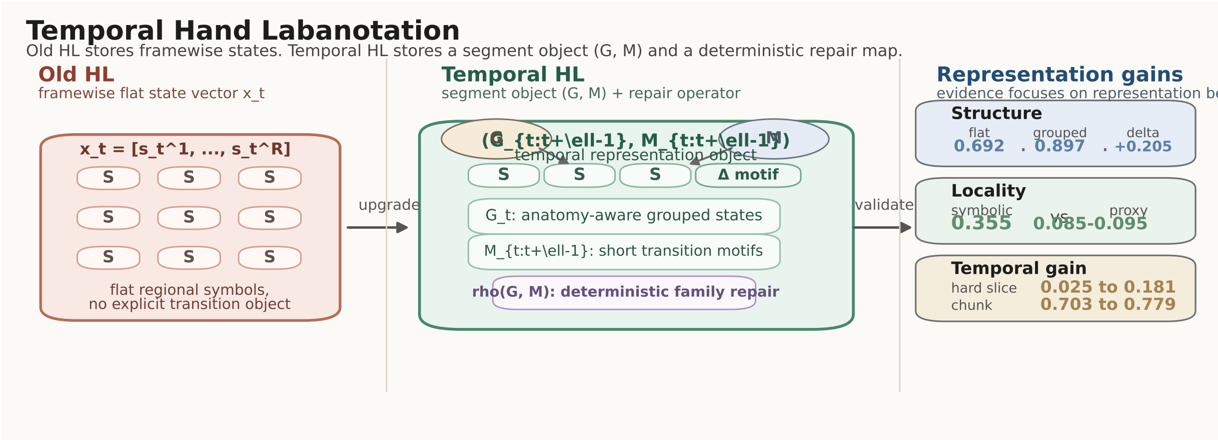

Figure 1: Overview of Temporal HL. Old HL stores each frame as a flat regional state vector. Temporal HL stores motion with grouped symbolic states plus short transition motifs, then applies deterministic family repair before downstream use. We evaluate the resulting representation through structure, locality, and interaction-conditioned preservation under matched downstream operators.

tif is the ordered symbolic change pattern attached to that run, not a latent hidden state used only during recognition.

Family repair. Grouped states and motifs are the representation itself. We then apply a family-level repair operator $\rho(G, M)$ that corrects residual within-family inconsistencies while preserving symbolic intent. The operator acts only inside anatomy-aligned families: it reconciles near-confusions among symbols that already share the same family context, while leaving other families and the target transition label unchanged. In this sense, $\rho(G, M)$ is a deterministic projection back to a family-consistent symbolic object, not a third learned component.

Temporal Structure in the Representation

Our design is driven by a representation question rather than a recognition question. Simply attaching a stronger temporal encoder to framewise HL is not the story we want to test. The representation itself must expose motion units that exist before the encoder. We therefore organize Temporal HL into three nested levels of symbolic granularity.

- State level: old-HL-style regional symbols remain available through the grouped state component G_t.
- Grouped level: these symbols are re-indexed into anatomy-aligned families rather than a single flat vector.
- Chunk level: the motif component $M_{t:t+\ell-1}$ captures the smallest temporal unit that still changes downstream preservation outcomes on the difficult interaction slices.

This hierarchy reveals where temporal headroom remains. On our interaction-heavy slices, sequence-level routing is too coarse, while chunk- or run-level symbolic units still retain useful signal.

Downstream Operators

The representation is validated through three downstream operators. They act on the defined object (G, M) rather than redefining the representation. This choice is deliberate. If the only evaluation were framewise recognition, then a stronger temporal encoder could improve the score even when the notation object itself had not become more editable or more reusable. We therefore test the representation on the operations it is supposed to support.

Local symbolic intervention. Given a target field such as interaction approach or right-hand opening, we ask whether a symbolic intervention can change only the requested field while preserving the others. This gives a clean locality metric.

Transfer. For hand-motion targets, donor frames alone are insufficient. We therefore use donor pairs or short chunks that already realize the requested transition. This evaluation tests whether temporal symbolic units transfer more reliably than framewise matches.

Interaction-conditioned preservation. The most difficult slices occur when a right-hand target motion must be preserved under interacting two-hand context. Here the evaluation uses a joint score:

$$\text{joint score} = \text{right grouped match} \times \text{left preserve}.$$

The right hand must match the grouped target motif, and the opposite hand must remain preserved.

Representation Evaluation Protocol

The downstream protocol has two stages. First, a support pool is formed from the validation split. For easy slices this can use strict donor matching; for hard right-hand interaction slices it requires corrected relaxed support so that the target-hand donor and preserve-side donor can be chosen more flexibly. Second, the selected support item is applied either as a local symbolic rewrite or as a transition-conditioned chunk transfer. This separation matters empirically: support relaxation accounts for the first major jump on hard slices, while task-specific preserve-side repair and chunk transfer provide later gains. We keep this support bank on validation rather than test data so that the final test evaluation remains strictly held out from support construction. Across comparisons, the downstream operator class is held fixed and only the underlying representation changes, so improvements cannot be explained by giving Temporal HL a privileged intervention mechanism. This is the minimal fairness condition for a representation paper: editability, preservation, and transfer should be attributed to the notation object rather than to a stronger downstream solver.

Experimental Setup

We reuse the HL annotation family introduced by Li et al. (2024), derived from InterHand2.6M and FreiHAND (Moon et al. 2020; Zimmermann et al. 2019). Our experiments operate on prepared temporal symbolic splits: a large train split for symbolic pretraining, a val split for support construction and model selection, and a test split for final evaluation. The exported benchmark contains 967/11/23 motion sequences and 12,460/2,739/5,218 frames in train/val/test, respectively. The val and test splits are held out at the sequence level and also introduce unseen subjects relative to train (subject 9 for val; subjects 10 and 11 for test). We evaluate three operator families on these held-out splits and additionally report hard right-hand interaction slices and their feasible interaction-rich subsets, which isolate the true two-hand preservation difficulty from frames where the opposite hand is effectively absent.

The hard slices are defined by requested right-hand motion changes under interaction-heavy contexts. A frame is considered feasible if the opposite-hand content is actually present and therefore preservable. This distinction is necessary: without it, the full-slice score is dominated by structurally absent-opposite-hand mass and understates the quality of the representation on the cases that matter.

We compare Temporal HL against four relevant alternatives: old HL, a flat symbolic variant without grouped temporal factors, learned-token proxies derived from continuous frame embeddings, and learned-token proxies derived from semantic frame embeddings. The intervention studies further compare strict support, corrected support relaxation, feasible-subset repair, and chunk transfer under the same held-out support protocol. Our main metrics are mean sequence accuracy, weighted symbolic locality under controlled edits, and joint interaction preservation on hard two-hand slices.

The learned sequence models use three seeds $\{0, 1, 2\}$ and report mean sequence accuracy. The temporal symbolic learners use window size 32, stride 16, hidden dimension 128, and 20+20 pretraining/finetuning epochs. The matched learned-token controls use the stronger schedule in our benchmark: token codebook size 64, maximum codebook frames 100,000, pretraining window span 186, step 96, and 120+120 epochs. Validation selects the learned-control hyperparameters. Chunk transfer uses the best corrected feasible-slice setting, with opening gains stable across chunk lengths 2–4. Symbolic intervention is deterministic once the split and support bank are fixed. For the hard interaction frontier, we additionally report bootstrap confidence intervals and paired permutation tests over evaluation frames. All compared representations share the same split family, held-out support protocol, and matched learned-token controls. The question is therefore not whether a larger temporal model can solve the task, but whether the notation itself yields better symbolic structure under matched operator strength.

Results

Main Results

Table 1 gives the paper's central verdict. Across all three blocks, Temporal HL is stronger precisely where the evaluation depends on explicit symbolic structure rather than on framewise matching alone. Table 2, the paper's only ablation table, then isolates which temporal channels are responsible for those gains. Taken together, these tests answer the representation question directly: structure tests ask whether the notation organizes state better, locality tests ask whether edits stay local, and interaction-preservation tests ask whether the same object can preserve motion under downstream use. Where applicable, the symbol inventory, operator class, and support protocol are held fixed, so the gains are evidence about the representation rather than about a privileged solver. The same pattern holds on unseen-subject held-out splits rather than only on in-distribution motion fragments.

Structure and Locality

The first block establishes the representation claim directly. Grouping the state factors raises sequence accuracy from 0.6923 to 0.8974, a gain of +0.2051 without changing the symbol inventory, so the improvement comes from how the notation organizes motion state rather than from enlarging the vocabulary. The same block shows a large locality margin: weighted symbolic locality reaches 0.3553, while matched semantic and continuous learned-token proxies remain at 0.0945 and 0.0848. When an edit must change one field while preserving the rest, the symbolic structure isolates the requested change far better than proxy token spaces trained only for generic separability.

Hard Interaction Preservation

The hardest setting is right-hand motion preservation under interaction, and it is also where frame-only HL

Table 1: Main results for Temporal HL as a symbolic representation. All deltas are absolute improvements under the same held-out protocol. The table follows the claim order of the paper: better symbolic structure, cleaner local editing, and stronger interaction-conditioned preservation and transfer. Where applicable, the symbol inventory and downstream operator are held fixed.

A. Structure and local editability				
Audit	Compared against	Base	Ours	Gain
Sequence accuracy	flat HL-style symbolic sequence	0.6923	0.8974	+0.2051
Weighted locality	semantic-token proxy	0.0945	0.3553	+0.2608
Weighted locality	continuous-token proxy	0.0848	0.3553	+0.2705
B. Hard interaction-conditioned preservation				
Setting	Compared against	Close	Open	Gain
Strict full slice	no support correction	0.0250	0.0533	baseline
Corrected support relaxation	strict full slice	0.1807	0.1812	+0.1556, +0.1279
Feasible two-hand subset	corrected full slice	0.6392	0.6497	+0.4585, +0.4685
Preserve-side repair	feasible two-hand subset	0.6582	0.6497	+0.0190, unchanged
C. Broader feasible chunk transfer				
Operator	Compared against	Close	Open	Gain
Fixed-edge transfer baseline	single donor edge match	0.7025	0.6306	baseline
Chunk transfer	fixed-edge transfer baseline	0.7785	0.7197	+0.0760, +0.0891

is least adequate. Corrected support relaxation already raises joint success to 0.1807 for closing and 0.1812 for opening. Once evaluation is restricted to genuinely feasible two-hand cases, the same representation reaches 0.6392/0.6497, and preserve-side repair further lifts closing to 0.6582 while maintaining 0.6497 on opening. The gain only appears once target motion and opposite-hand preservation are evaluated together rather than matched independently.

These gains are statistically stable. On the full hard slice, corrected support relaxation improves closing by +0.156 (95% CI $[0.127, 0.186]$) and opening by +0.128 (CI $[0.101, 0.156]$), with paired permutation tests below 10^{-4} in both cases. On the feasible interaction subset, preserve-side repair adds another +0.184 on closing ($[0.108, 0.259]$, $p = 5 \times 10^{-5}$) and +0.166 on opening ($[0.108, 0.223]$, $p < 10^{-4}$).

These results show that temporal symbolic quality becomes visible only once the evaluation respects the two-hand constraint. After target matching is handled correctly, the remaining headroom lies in preserve-side consistency, so the later gains come from repair and chunk transfer rather than donor selection alone.

Chunk-Level Transfer

Chunk transfer gives the clearest temporal gain on broader corrected feasible slices: Table 1 shows improvements from 0.7025 to 0.7785 on closing and from 0.6306 to 0.7197 on opening. The opening gain is stable across chunk lengths 2, 3, and 4, which indicates that the benefit comes from short local transition structure rather than long-horizon modeling. Table 2 aligns with this result within the learned sequence-model family: the best formulation is symbolic state plus short transition channels, while heavier duration and coordination channels do not match it and the learned-token control remains well below it. The paired slice comparison is also clean: on broader feasible closing, chunk transfer wins on 12 frames, loses on 0, and ties on 146; on opening, it wins on 14, loses on 0, and ties on 143.

Table 2: Representation ablation within the learned sequence-model family. The best formulation is the minimal temporal upgrade: symbolic state plus short transition motifs. Deltas are reported against the best row.

Formulation	Seq. acc.	Δ vs best
State + transition	0.9231	best
State + transition + duration	0.8205	-0.1026
State only	0.7949	-0.1282
State + transition + coordination	0.7436	-0.1795
Learned-token control (64)	0.6923	-0.2308

Temporal Granularity Analysis

The remaining gap localizes where stronger symbolic supervision is still needed. Interaction preservation improves sharply once support is corrected and evaluation focuses on feasible two-hand cases, while chunk transfer adds a second gain layer on broader feasible slices. The residual difficulty is now concentrated on preserve-side consistency in interaction-heavy cases, not on whether temporal structure belongs in the representation.

Discussion and Limitations

Temporal HL upgrades HL into a hand-motion representation with explicit short-range temporal structure. Relative to frame-only HL, it better preserves structure, supports local edits, and transfers motion under interaction. The strongest gains appear exactly where the evaluation depends on temporal symbolic content.

The upgrade is also minimal in the right way. Temporal HL preserves the original HL inventory, reorganizes it into anatomy-aligned state factors, and adds short transition motifs only where framewise notation is insufficient. This keeps the notation compatible with HL while changing what downstream symbolic operators can do.

This is also why we emphasize representation-level evidence rather than recognition-only gains. For a notation paper, the strongest claim is not that a decoder can classify frames more accurately, but that the symbolic object supports the operations for which a notation is actually used.

We also keep the scope of that claim narrow. The paper does not argue that these tests exhaust every downstream use of hand notation; it argues that they cover the core operations a symbolic hand-motion representation should support before broader generalization is considered.

The current frontier is preserve-side consistency under dense interaction. The remaining headroom lies at run or chunk granularity, where simple symbolic selectors do not yet recover all feasible context. A natural next step is interaction-aware temporal supervision. We keep the present study within the HL-derived benchmark so that the gains reported here remain attributable to the representation itself.

Conclusion

Temporal HL is a symbolic representation for hand motion with short transition structure inside the notation. Relative to a frame-only formulation, it preserves structure more reliably, supports local editing, and improves motion transfer under interaction. Time-local transition structure belongs to the representation itself, not to an optional decoder-side refinement. The resulting notation object supports retrieval, intervention, reuse, and more direct symbolic interfaces for downstream hand-motion systems.

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